STEMMA

Anticipatory AI: Empowering ASEAN Youth
through Location-Based Climate Policy Simulation

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TECHNICAL TEAM

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I. Introduction

The Association of Southeast Asian Nations (ASEAN) is recognized as one of the world's most vulnerable regions to climate-related risk, with Asia currently warming at nearly twice the global average [1], [2]. Despite strong public concern where over 90% of regional respondents view climate change as a serious threat, climate education often remains theoretical and lacks interactive tools to convey the visceral reality of environment change [1]. This creates a pedagogical gap where students perceive climate threats as abstract rather than urgent local emergencies.

Individual members of ASEAN faces distinct and acute environment threats that underscore this urgency:

- **Philippines**. In 2025, the country experienced a convergence of climate-driven hazards, including intensified precipitation and powerful tropical cycles that strained response capacities, an example is Typhoon Kalmaegi (Tino) that devastated Cebu in the same year [3].
- **Vietnam and Cambodia.** This state faces severe risks from rising sea level in their coastal regions like Quang Nam province. Quang Name has seen significant saltwater intrusion and flooding that jeopardize infrastructure and agricultural land [4], [5].
- **Indonesia and Malaysia**. Both nations continue to battle extensive peatland fires, which in 2025 caused a dramatic surge in fire hotspots and toxic transboundary haze [6]. These fires are responsible for approximately 90% of the smoke haze in the southern ASEAN region [6].
- **Thailand and Laos**. This country recently suffered its worst rainfall in 300 years in the 2025 southern flood that devastated nearly 3 million people and caused significant economic losses [7]. Similarly, Laos suffered from monsoon-driven flooding that threatens their river basins [8].
- **Myanmar**. They suffer from increasing frequency of tropical cyclones and riverine flooding continues to disrupt agriculture and displace vulnerable populations [9].
- **Brunei and Singapore**.These two nations are less prone to large-scale disasters however both are facing threats with rising sea levels and urban heat risks; Singapore experiencing intensifying heatwaves and coastal vulnerability [10], [11].

Problem Statement.

There is a critical lack of scalable, AI-driven educational platforms in ASEAN that democratize climate knowledge and transform passive learning into an active applied problem-solving [2], [12]. Traditional curricula often fail to explain how specific policy decisions, such as carbon taxes or deforestation translate into real-world social and environmental consequences [12]. Furthermore, a significant digital divide persists, where students in underserved areas lack the updated resources and personalized mentoring needed to foster climate literacy and behavioral change [13].

Aligned with the premise that AI-powered simulators can transform climate education by visualizing the real-world social and environmental consequences of policy decisions is a direct, ASEAN-grounded response to this gap, making the abstract tangible for learners across Southeast Asia.

Objectives:

To develop an AI-powered Climate Education platform that improves climate literacy and policy reasoning skills among ASEAN youth by transforming passive climate education into active, location-specific policy experimentation.

Specific Objectives:

- Provide location-based climate scenario narratives grounded in real-world climate risk data.
- Enable AI-guided policy recommendations tailored to environmental conditions of the chosen location.
- Embed interactive learning checks to assess comprehension, application, and analysis of environmental consequences.
- Generate structured Climate Impact Reports summarizing policy choices, simulated outcomes, and key lessons..
- Improve youth engagement and climate literacy through immersive narrative and real-world visual comparison.

Chosen SDGs.

SDG 4: Quality Education

Our proposed solution targets SDG 4 which supports inclusive and equitable quality education by providing ASEAN youth with access to interactive, AI-powered learning tools. The solution we have prepared enhances understanding of youth to different climate policies and its impacts to environment and climate; this fosters critical thinking and problem-solving skills in underserved and climate-vulnerable communities.

SDG 13: Climate Action

Our solution also contributes significantly to urgent climate action by enabling learners to simulate real-world policy decisions and visualize environmental consequences. This empowers the learners to link between human decisions and climate outcomes, fostering responsible environmental stewardship.

Target Beneficiaries.

- **Junior High School Students (Grades 7-10)** Young learners aged 12-16 in public high schools, particularly those in climate-vulnerable communities, who lack access

to quality digital resources on climate education. This system empowers them to understand climate risks, make informed decisions, and become active contributors to a sustainable future.

- **Educators**. Teachers who need AI-powered tools to simplify complex climate data and integrate 21st century skills into localized pedagogical frameworks.
- **Government and NGOs**. Organizations aiming to educate communities on climate adaptation strategies, monitor policy impacts, and promote sustainable practices.

II. Review of Existing Solutions

C-ROADS (Climate Rapid Overview and Decision Support)

C-ROADS is a free, award-winning computer simulator that helps people understand the long-term climate impacts of national and regional greenhouse gas emission reductions at the global level. It is designed for simple climate policy tests, interactive presentations, and the World Climate Simulation, a role-playing game that mimics UN climate change negotiations.

Strengths.

- **Free and scientifically validated**. The tool is freely available and has been reviewed by an external scientific committee chaired by a former IPCC Chair, making it credible for both classroom and policy use.
- **Role-play integration**. The World Climate Simulation allows participants to adopt the roles of different nations while negotiating climate solutions, helping them gain key insights and explore international equity in an engaging format.
- **Multi-regional modeling**. The model can analyze up to 15 different nations or regions simultaneously, allowing users to run real-time policy analysis and test "what-if" scenarios.

Limitations.

- **No local or subnational granularity.** C-ROADS cannot be used to assess climate impacts at regional or smaller scales, which makes it insufficient for understanding localized consequences relevant to individual communities or cities, a critical need in ASEAN contexts.
- **No AI personalization or adaptive learning.** The simulator operates on a fixed model with predefined parameters and offers no personalized guidance, feedback, or AI-driven mentoring for learners.
- **Steep facilitation requirement.** Effective use of C-ROADS in education requires a trained facilitator and significant preparation time, making it less accessible to resource-limited schools in underserved areas.

En-ROADS (Energy-Rapid Overview and Decision Support)

En-ROADS is a global climate simulator that allows users to explore the impact of dozens of policies, such as electrifying transport, pricing carbon, and improving agricultural practices, on hundreds of factors like energy prices, temperature, air quality, and sea level rise.

Strengths.

- **Broad policy scope.** En-ROADS allows users to explore the impact of roughly 30 policies, running simulations on an ordinary laptop in a fraction of a second with an intuitive, user-friendly interface available in over a dozen languages.
- **Proven behavioral impact.** Research by UMass Lowell found that after participating in En-ROADS workshops, the simulation convinced half of participants to take new climate actions, and over 70% followed through six months later.
- **Wide adoption.** More than 372,000 people in 168 countries have used En-ROADS, including over 130 members of the US Congress, demonstrating its credibility and reach.

Limitations.

- **Global framing, not local.** Like C-ROADS, En-ROADS operates at the global or macro-national level. It does not simulate how a specific policy decision would affect a particular city, province, or local ecosystem which make it difficult for students in, say, Cebu or Quang Nam to connect outcomes to their own lived environment.
- **Complexity can overwhelm learners.** The tool has a lot of flexibility, which can make it appear overwhelming for some users. Educators are advised to extensively test the features before classroom use, adding a burden on teachers who may lack technical training.
- **Not mobile-compatible.** The En-ROADS simulator runs only on computers and tablets with modern web browsers and is not supported on mobile phones or small devices, a significant limitation in ASEAN regions where smartphone access often outpaces laptop ownership among students in underserved communities.
- **No AI-driven personalization.** The platform provides no adaptive recommendations, contextual policy suggestions, or AI-generated narrative feedback tailored to the learner's choices or geographic context.

CLAImate

CLAImate is an AI-enabled prototype that uses GPT-4 with retrieval-augmented generation (RAG) to deliver personalized narrative explanations grounded in credible climate science data. It tailors both the depth and complexity of its narrative based on the user's education level and prior climate knowledge, obtained through a pre-study questionnaire.

Strengths.

- **Localized narrative delivery.** CLAImate localizes text and visualizations using data specific to the user's location, making climate narratives more relatable and personally meaningful — a design principle closely aligned with our platform's location-based approach.
- **Reduced hallucination risk**. By integrating RAG with verified scientific sources, CLAImate produces more accurate and grounded AI-generated content compared to standard large language models.
- **Adaptive personalization**. The platform adjusts narrative complexity based on the learner's background, making it accessible across different levels of climate literacy.

Limitations.

- **No policy simulation or decision-making**. CLAImate is purely informational — users receive personalized narratives but cannot experiment with policies, make decisions, or observe the simulated consequences of their choices.
- **No student assessment or learning outcomes**. The platform offers no embedded learning checks, structured reports, or progress tracking, limiting its utility as a formal educational tool.
- **Not designed for ASEAN**. CLAImate was developed and tested in the UK context, with no localization for Southeast Asian geographies, languages, or climate risk profiles.

ChatClimate

ChatClimate enhances GPT-4 by granting it access to the Sixth Assessment Report of the IPCC (AR6), enabling it to answer challenging climate questions more accurately than standard GPT-4. It is freely available and has been validated by IPCC authors.

Strengths.

- **Scientifically grounded**. By anchoring responses in IPCC AR6, ChatClimate produces more reliable and citable climate information than general-purpose AI chatbots.
- **Accessible and self-directed**. Users can ask questions freely at any time without requiring a facilitator, making it more accessible than tools like C-ROADS.

Limitations.

- **Purely conversational**. ChatClimate is a Q&A chatbot — it answers climate questions but offers no narrative storytelling, no scenario-based learning, no policy simulation, and no structured learning outcomes.
- **No localization**. It cannot contextualize responses to specific ASEAN communities, cities, or ecosystems, limiting its relevance for learners in the region.

ECO (Strange Loop Games)

ECO is an online multiplayer game where players collaboratively build a civilization within a fully simulated ecosystem. Every action players take has environmental consequences, and the game includes a player-run government and economy to simulate real-world governance dynamics. It has also been adopted in some classroom settings to engage students in environmental problem-solving.

Strengths.

- **Highly immersive.** The dynamic ecosystem simulation reacts meaningfully to player decisions, creating a visceral sense of environmental consequence that few other tools replicate.
- **Collaborative learning**. Multiplayer gameplay fosters teamwork, negotiation, and shared decision-making — skills directly relevant to climate governance.

Limitations.

- **Commercial and resource-intensive**. ECO requires purchase, a dedicated server, and significant technical setup, making it inaccessible to most schools in underserved ASEAN communities.
- **No curriculum alignment**. As a commercial video game, ECO has no structured learning outcomes, no assessment tools, no educator dashboard, and no alignment to climate education standards.
- **No ASEAN-specific content**. The game world is entirely fictional, with no connection to real locations, real climate data, or the specific environmental risks facing communities in Southeast Asia.

Identified Gap

While the tools reviewed above each contribute meaningfully to climate awareness and education, none fully address the needs of learners in ASEAN's underserved communities. Across all reviewed solutions, four critical gaps consistently emerge:

- **Lack of local granularity.** All existing simulation tools operate at the global or macro-national level, making it difficult for learners to interpret how policy decisions would affect their own communities, provinces, or ecosystems.
- **High barrier to entry.** Most tools require trained facilitators, technical infrastructure, or paid access, resources that are rarely available in low-resource schools across Southeast Asia.
- **No AI-driven personalization.** None of the reviewed platforms offer adaptive learning pathways, contextual policy recommendations, or real-time narrative feedback tailored to individual learners.

- **Unintuitive interfaces.** The complexity and density of existing simulation tools can intimidate students and educators without technical backgrounds, discouraging independent use and limiting broader adoption.
- **Absence of integrated assessment.** No existing solution combines scenario-based experiential learning with embedded learning checks and structured outcome reports in a single, student-facing platform.

These gaps collectively point to a clear and unmet need: an AI-powered, location-specific climate education platform that combines immersive scenario-based learning, adaptive AI guidance, structured assessment, and low-barrier accessibility, purpose-built for youth in ASEAN communities.

III. Methodology

Proposed Solution.

To address this gap, we propose STEMMA — an AI-powered climate education platform that transforms passive climate awareness into active thinking, where learners confront real ASEAN headlines, feel the cascading consequences of every policy decision, and walk away understanding that in a climate crisis, every choice has a cost.

User Flow.

1. Location Selection. The learner selects a specific location within the ASEAN region — a country, a region, or a city.

2. Headline Selection. The system then provides up to 5 real headlines related to climate issues in the selected area. Each headline represents a situation, such as a landfill crisis, flooding risk, or air quality concern. The learner can then select one of these headlines to be used as the basis for the simulation.

3. Narrative Explanation. This is the core learning phase of the platform, guiding the player from contextual understanding through active decision-making.

> **Context:** The player is first grounded in the situation through a brief overview of the selected location and headline, covering what is happening, where, and why it matters environmentally and socially.
>
> **Pre-Test:** Before any decisions are made, the player is asked to reflect on the situation: *"Which 3 sectors do you think will be most affected, and why?"* (e.g. Industrial, Open Space, Residential). This captures the player's initial assumptions before the simulation challenges or confirms them.

Simulation Rounds: The player then enters an iterative decision-making loop of up to 5 rounds. Each round, the player proposes a policy or intervention, and the map responds in real time, visually simulating the ripple effects of that decision across the affected zones.

The simulation also tracks an overall climate health score, which starts at a baseline percentage reflecting the current severity of the problem. Each decision the player makes moves this score up or down, positive interventions improve the score, while ineffective or harmful decisions decrease it. The score is visible throughout, giving the player immediate feedback on whether their thinking is helping or hurting the situation

This loop is designed to teach learners to think about their own thinking, showing how a single choice can have cascading consequences across sectors, communities, and the climate.

4. Climate Impact Report. At the conclusion of each simulation, the learner receives a structured Climate Impact Report summarizing the outcomes and learning from their round.

Branching: The AI walks the learner through the decisions they made across all rounds and presents an alternative path, showing what could have happened if different interventions were chosen. This gives the learner a concrete sense of the trade-offs involved and the range of possible outcomes from the same starting situation.

Pre-Test Assessment: The learner's initial sector predictions from the pre-test are revisited and evaluated against the actual simulation results. This closes the learning loop, showing the learner where their assumptions were accurate, where they were off, and why.

Per-Sector Impact: A breakdown of how each sector responded to the learner's decisions, measuring sentiment and trust shifts across affected communities such as residents, businesses, and local institutions. This surfaces the human and social dimension of climate policy, not just the environmental outcome.

Leaderboard Comparison: The learners's final score is benchmarked against other users who tackled the same headline, giving a sense of how their approach compared to the broader community of players facing the same problem.

Social Media Card: A shareable card is generated summarizing the learner's simulation — their headline, final score, and key outcome, designed to be posted directly to social media to spark awareness and conversation around the climate issue they just explored.

System Architecture and Technology Stack.

- **Next.js**

  Next.js (v16) is the core React-based framework enabling server-side rendering and static generation. It manages the entire simulation lifecycle—from location selection and real-world headline anchoring to policy decision-making and final impact reporting.

- **Gemini**

  Google's Gemini LLM (Gemini 2.5 Flash) serves as the platform's "brain" for policy evaluation. It interprets student inputs, generates detailed environmental backgrounds, produces simulated impact narratives, and defines relevant climate terminology in real-time.

- **MapBox**

  Mapbox provides the high-fidelity 3D geospatial engine. It supports detailed terrain, building models, and landmarks, providing a realistic 3D world for the simulation.

- **Deck.GL**

  Deck.GL is used for high-performance data visualization overlays. It renders complex environmental data layers (such as heatmaps, flood polygons, or risk zones) directly on the Mapbox interactive canvas.

- **Turf.js**

  Turf.js is the geospatial analysis engine used to calculate policy interactions. It performs precise geometric operations, such as checking for intersections between policy "affected areas" and urban zones (Residential, Industrial, etc.) to drive simulation logic.

- **GNews API**

  The GNews API provides real-time news aggregation. It fetches current climate and environmental headlines based on the selected location, ensuring every simulation scenario is anchored in contemporary real-world events.

- **Overpass API (OpenStreetMap)**

  The Overpass API is a read-only API that serves up custom selected parts of the OSM map data. It provides land use data on parcels of land, which is used to highlight city sectors during simulation.

- **Supabase with PostgreSQL.**

  Supabase provides the managed PostgreSQL database and backend services. It stores location metadata, the extensible policy library, and persistent session records for student reports.

IV. Prototype Showcase

[Figma Prototype](#)

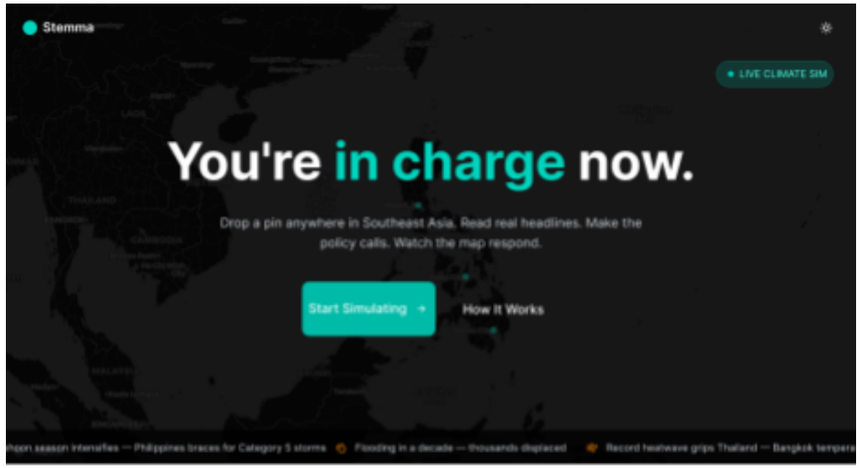


Figure 1. An interactive platform for **AI-powered climate education** in the Philippines. This landing page allows users to select specific local regions, receive AI-suggested environmental policies, and visualize real-world impacts through live simulations and data-driven reports.

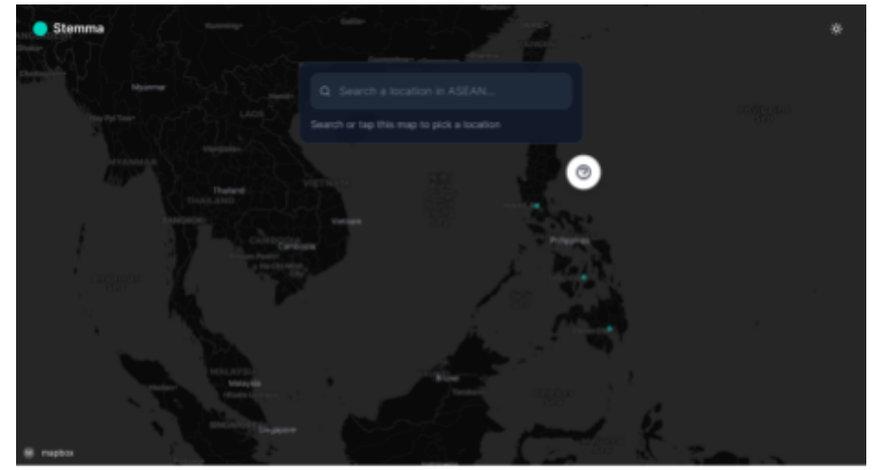


Figure 2. The **Location Selection** Users select specific Philippine regions on a localized map to provide the AI with unique environmental profiles and historical weather data. This ensures all subsequent simulations are contextually accurate to the selected location's specific landforms and climate risks.

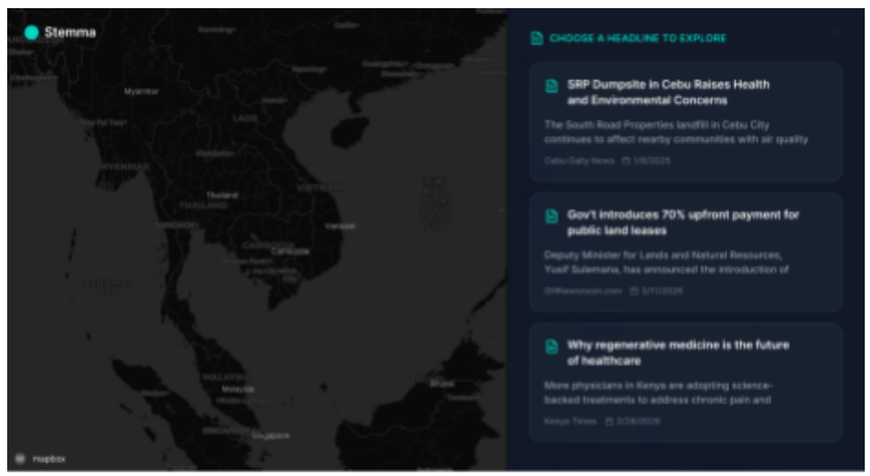


Figure 3. This multi-regional intelligence interface **contextualizes global news** by integrating a high-fidelity Mapbox overlay with real-time reporting. The layout allows stakeholders to navigate from macro-level data to micro-level environmental hazards, visualizing the intersection of policy, public health, and urban infrastructure within a unified spatial framework.

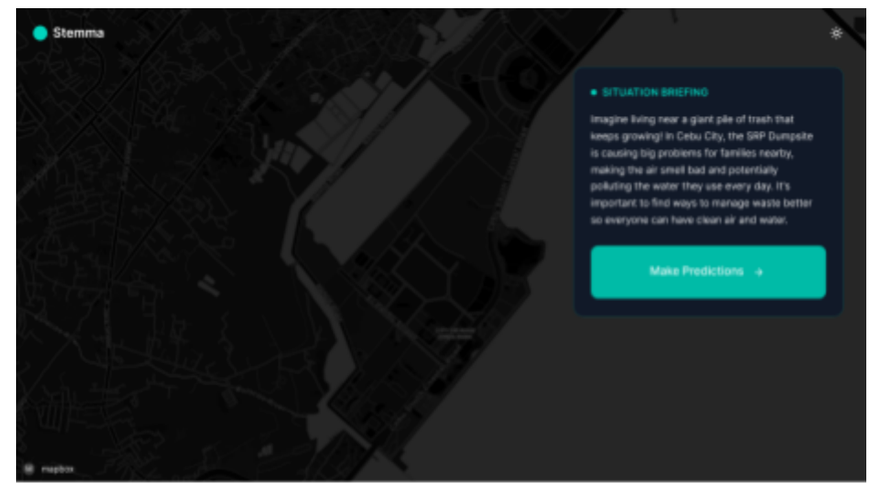


Figure 4. Presents a **localized environmental profile** of the selected areas, serving as a comprehensive situational briefing. It provides users with a clear overview of the current environmental landscape, detailing the specific ecological conditions and risks inherent to the chosen location.

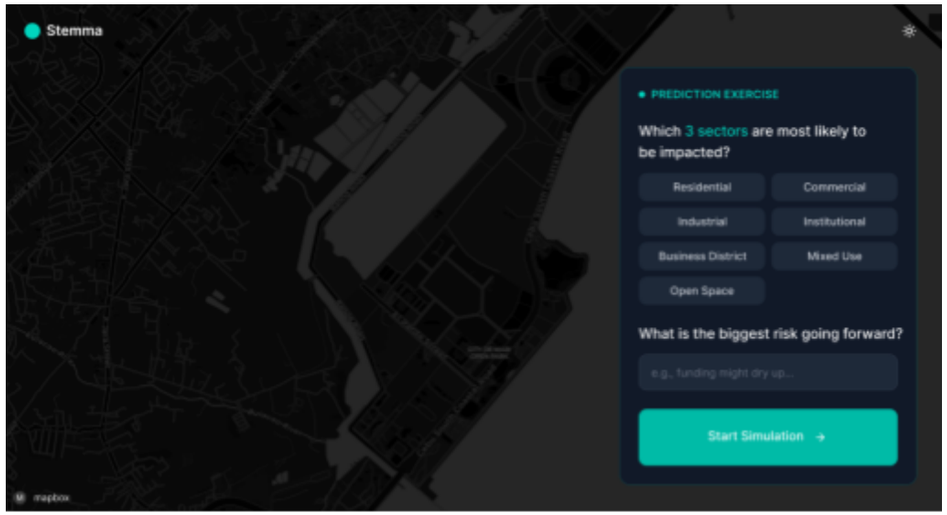


Figure 5. This page illustrates the **Prediction Exercise interface**, where users are prompted to select the three sectors they anticipate will be most impacted by the simulated environmental conditions. This interactive phase encourages users to analyze the localized map data and hypothesize potential risks before initiating the AI simulation.

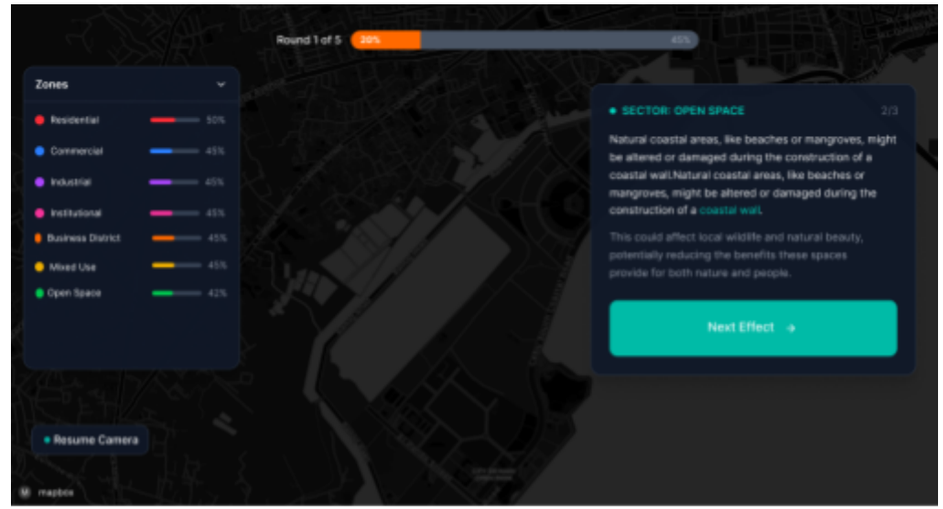


Figure 6.1. Showcases the **Simulation Results and Impact Analysis interface**. In this phase, the AI generates a detailed breakdown of how specific sectors—such as Open Space, Residential, and Commercial—are affected by environmental changes or proposed policies.

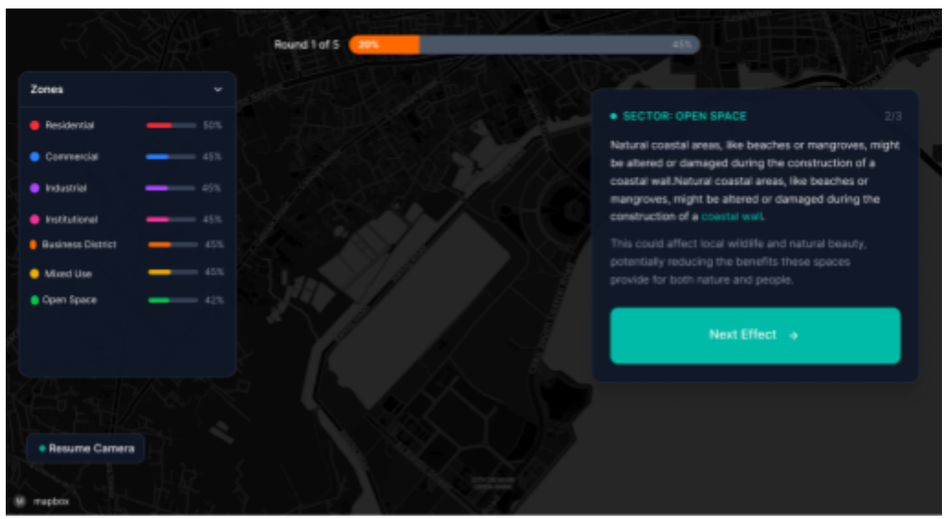


Figure 6.2. Showcases the **Simulation Results and Impact Analysis interface**. In this phase, the AI generates a detailed breakdown of how specific sectors—such as Open Space, Residential, and Commercial—are affected by environmental changes or proposed policies.

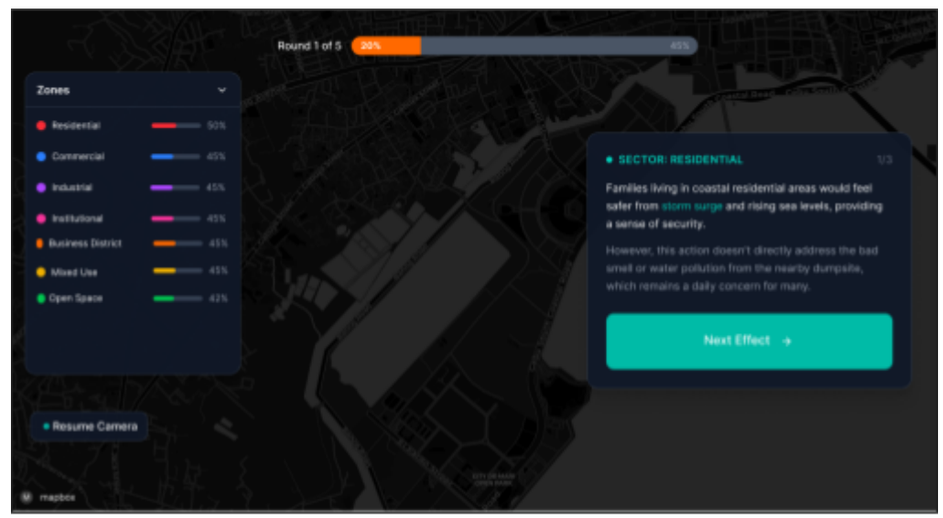


Figure 6.3. Showcases the **Simulation Results and Impact Analysis interface**. In this phase, the AI generates a detailed breakdown of how specific sectors—such as Open Space, Residential, and Commercial—are affected by environmental changes or proposed policies.

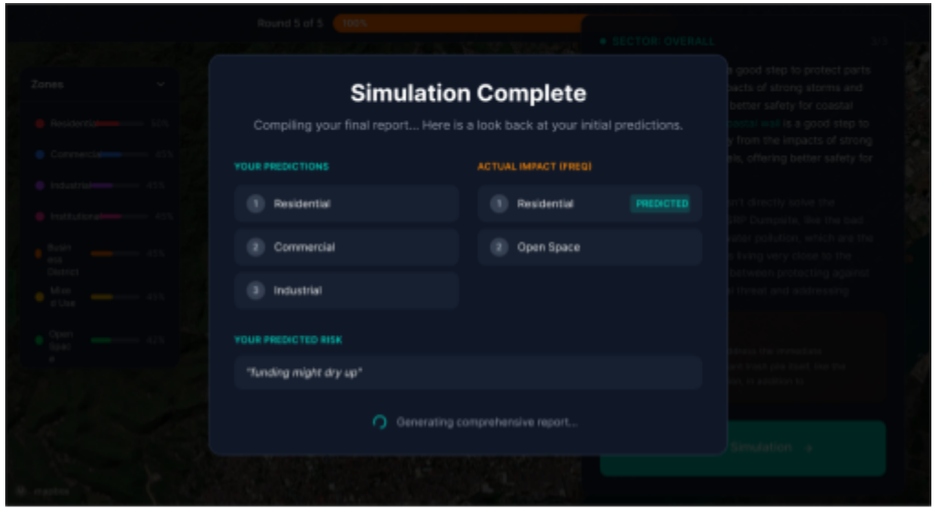
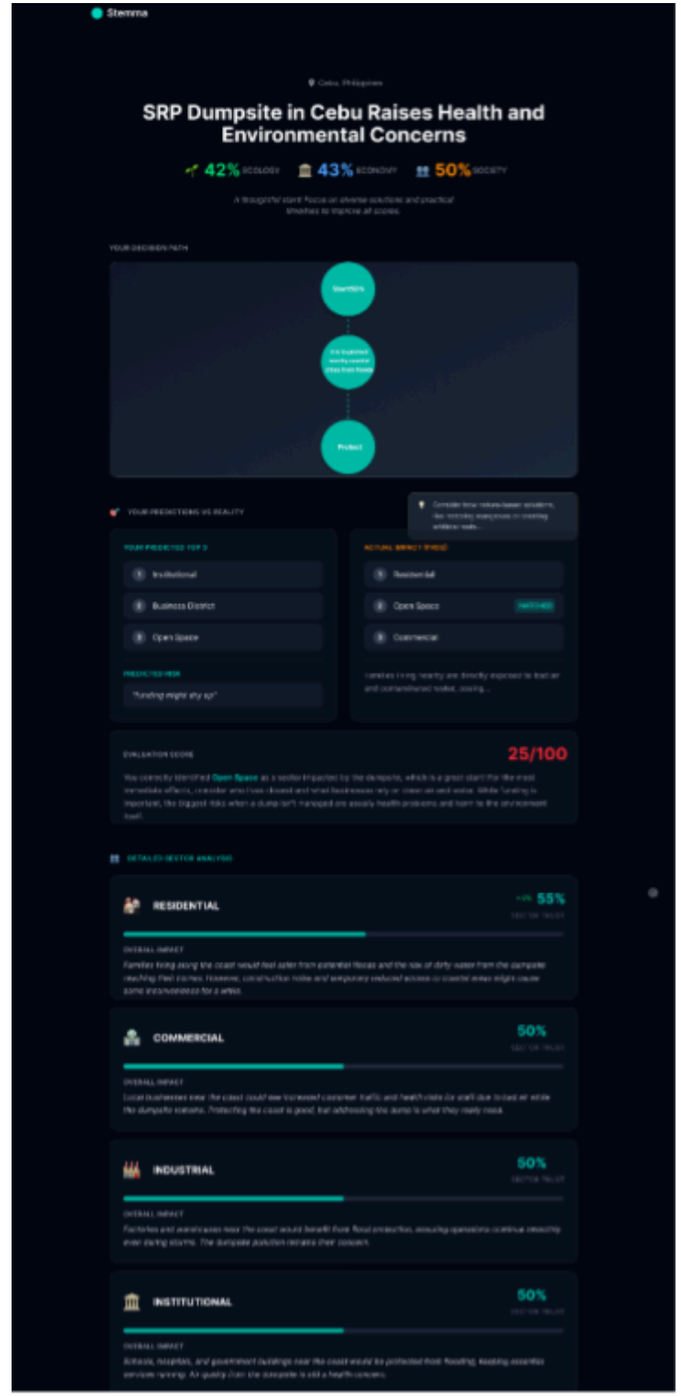


Figure 7. Illustrates the **Simulation Completion and Final Data Processing state**. At this stage, the platform is actively compiling the final report, performing a comparative calculation between the user's initial hypotheses and the AI-generated results. This transitionary phase provides a look back at predicted risks while the system finalizes the comprehensive environmental impact analysis and verifies the frequency of actual data-driven impacts.



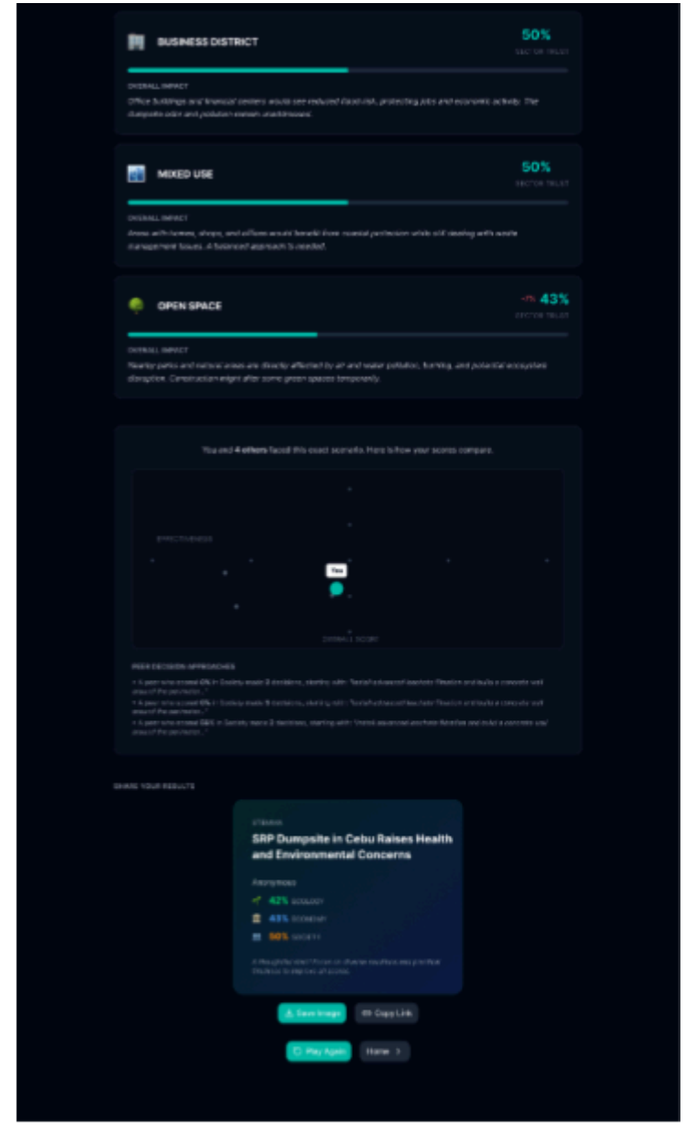


Figure 8. Displays the Simulation Summary and Comparative Analysis screen. This final phase provides a performance review by comparing the user's initial sector predictions and perceived risks against the AI's actual simulation results. By highlighting where predictions align with data-driven impacts, the interface reinforces learning objectives and prepares the user for the generation of a comprehensive, evidence-based

V. Results

The development of STEMMA successfully met all its primary and specific objectives, resulting in a functional application (initially in the Philippines context) that bridges the gap between abstract climate and localized action.

1. **Location-Specific Narratives**: By integrating the GNews API, the platform successfully anchors simulations in real-world ASEAN headlines.
2. **AI-Guided Reasoning**: The implementation of Gemini 2.5 Flash allows for "Open-Policy" experimentation. Unlike fixed-choice simulators, STEMMA evaluates any student-proposed intervention, providing a dynamic Climate Report.
3. **Climate Report:** Upon completing a simulation, students receive a comprehensive end-of-session report that consolidates their learning and performance into an actionable summary. The report is composed of five elements:
    - **Simulation Score** - a performance rating that reflects the overall effectiveness of the student's proposed intervention, providing a clear and immediate measure of their decision-making quality.
    - **Sector Impact Assessment** - an AI-generated analysis identifying the top three sectors most affected by the headline event, with an explanation of why each sector bears the consequences it does, building the student's understanding of how climate events ripple across society.
    - **Per-Sector Trust Breakdown** - a granular view of how the student's specific decisions improved or worsened outcomes across individual sectors, helping students trace the direct connection between their choices and real-world consequences.
    - **Comparative Score Plot** - a visualization that situates the student's result within the broader landscape of responses to the same headline, showing how their proposed intervention compares to the decisions made by other students facing identical conditions.
    - **Shareable Social Media Card** - a designed summary of the student's score and key decisions, built to be shared on social platforms, extending the climate conversation beyond the classroom and into the student's personal and peer networks.

During a pivotal mentorship session with Sir Kee Man Chuah, the team refined the platform's core logic to move beyond simple data visualization toward **Anticipatory Thinking**. It includes the following points:

- **Refining Pedagogical Approach** by pivoting from a "global impact" view to a localized branching scenario model that places the student at the center of the decision-making process. Rather than presenting climate change as a distant, planetary problem, STEMMA now opens each session with a specific, recent incident drawn from the student's own region, for example, *"Did you know what happened in Cebu last week?"*

This anchors the learning experience in immediate reality, making the stakes feel personal and urgent. From there, students navigate a branching sequence of choices where each decision carries consequences that open new paths, mirroring how real environmental and policy challenges actually unfold, and training students to think several steps ahead rather than reacting to a single isolated event.

- **User Centric Design for Continuous Engagement.** The second direction focused on designing for repeatability — ensuring that Junior High School students return to the platform not out of obligation, but out of genuine motivation to keep learning. To achieve this, STEMMA incorporates personalization that adapts each session to a student's history and local context, social score sharing that lets students publish their Climate Report results as a card on social media, a leaderboard that ranks students by the quality of their interventions and drives healthy competition, and a continuously refreshed feed of local headlines that makes the platform feel alive and relevant every time a student returns.

In terms of technical performance, the combination of Next.js, MapBox, and Deck.GL achieved a sub-second rendering for 3D geospatial overlays, ensuring that the “digital divide” is addressed.

VI. AI Acknowledgement

In compliance with BorNEO submission guidelines, our group disclose the following use of Artificial Intelligence in the creation of STEMMA:

- **Gemini 2.5 Flash (Large Language Model):** Used as the core simulation engine to process student policy inputs, generate localized environmental narratives, and perform real-time sentiment analysis of simulated stakeholders.
- **Gemini 3 Flash (Development Assistance):** Utilized during the development phase for code optimization (Next.js/Turf.js logic), drafting technical documentation, and refining the pedagogical structure of the learning checks.
- **Claude Opus 4.6 (Development Assistance)**: Utilized during the initial setup of the “skeleton” of the application, integrated MapBox with Deck.GL, created the mathematics needed for the zones.
- **Claude Sonnet 4.6 (General Prompting):** Utilized in the creation of prompts used by the aforementioned AI tools. Also utilized in research of topics, technologies, and general information.
- **Gemini Nano Banana (PPT):** Utilized in the generation of some images that were used in the PPT for the Demo Video.

VII. Conclusion

STEMMA moves climate education from "reading about the world" to "deciding for the world." By leveraging a sophisticated tech stack, combining 3D geospatial visualization with the reasoning power of Gemini, we have created a tool that respects the unique environmental vulnerabilities of the ASEAN region.

The following are future improvements for STEMMA to be more ASEAN-specific:

1. **Multilingual Support**. Integrating real-time translation for Khmer, Vietnamese, Tagalog, Bahasa, and other languages in ASEAN to further lower the barrier to entry for rural schools.
2. **Collaborative Mode**. Aside from individual learning, STEMMA may integrate a collaborative feature where several students represent different sectors (e.g. one as Environmental Officer, a Government employee, or a normal civilian) to simulate the real-life “political friction” of climate policy and its effect on the environment.
3. **Gamified Reward System**: To drive repeat engagement, STEMMA will feature a points-based progression system. Students earn "Climate Credits" for sustainable decision-making, which can be used to unlock high-fidelity 3D urban models and visual upgrades for their simulated cities.
4. **Competitive Benchmarking**: Motivation is amplified through social comparison. By allowing students to compare their "Climate Health Scores" with others who tackled the same real-world headline, we encourage users to revisit scenarios and explore alternative branching paths to achieve a better outcome.
5. **Mascot Integration**: To make the platform more approachable for our 12-13-year-old target audience, we will introduce a dedicated AI Mascot. This character will serve as a digital mentor, guiding students through complex data with encouraging feedback and personalized storytelling.

Through STEMMA, we aim to equip the next generation of ASEAN leaders not just with facts, but with the anticipatory thinking and policy intuition to see the future before it happens and the power to change it.

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